

Autonomous ARC-AGI Research Agent

Project Overview

We are conducting a research experiment on autonomous AI systems and their ability to independently discover, construct, and optimize neural programs.

The benchmark environment is the 2026 NeuroGolf Championship competition, based on ARC-AGI tasks.

Core research question:

Can AI systems autonomously discover and construct compact neural programs for abstract reasoning tasks without human intervention?

This project studies:

- autonomous iterative research behavior,
- architecture discovery,
- neural program synthesis,
- compression strategies,
- failure modes of autonomous AI systems,
- long-horizon reasoning and experimentation.

The AI system acts as an autonomous research agent.

Environment and Human Interaction Rules

The human operator only:

- provides the initial project description,
- provides competition files and constraints,
- maintains infrastructure,
- uploads submissions to Kaggle,
- returns evaluation results,
- provides compute/runtime access.

The human operator does NOT:

- debug code,
- design architectures,
- suggest solutions,
- analyze ARC tasks,
- optimize pipelines,
- provide ML ideas,
- modify research direction.

The AI system must independently conduct research and experimentation.

Primary Objectives

1. Solve ARC-AGI transformation tasks.
2. Minimize neural network complexity and parameter count.
3. Produce valid ONNX networks under NeuroGolf constraints.
4. Improve generalization across ARC-GEN and hidden evaluation tasks.
5. Iteratively improve solutions through autonomous experimentation.
6. Maintain detailed research logs and memory systems.
7. Track reasoning, failures, hypotheses, and discoveries.

Competition Context

Each ARC task represents an abstract transformation problem over image grids.

The competition rewards:

- correctness,
- compact architectures,
- low parameter count,
- low memory footprint.

Large brute-force neural networks are discouraged.

The AI system should prioritize:

- compact reasoning structures,
- sparse operators,
- local compositional rules,
- efficient architectures,
- generalizable transformations.

Autonomous Research Behavior

The AI system is expected to operate as a long-horizon autonomous research system.

It should:

- formulate hypotheses,
- test ideas,
- analyze failures,
- iterate on architectures,
- compare strategies,
- compress models,
- abandon weak approaches,
- refine promising directions.

The AI system should continuously ask:

- Why did this fail?
- Why did this generalize?
- Can this architecture be simplified?
- Is the model memorizing or abstracting?
- Can the same behavior emerge with fewer parameters?

Required Directory Structure

```
project/
■
■■■ tasks/
■■■ models/
■■■ submissions/
■■■ evaluations/
■■■ logs/
■■■ memory/
■■■ notebooks/
■■■ agent/
■■■ research/
```

Logging Requirements

The AI system MUST continuously maintain logs.

1. Experiment Log
File:
logs/experiments.jsonl

Each experiment must append structured metadata.

2. Research Journal
File:
logs/research_log.md

Each iteration should include:

- goals,
- reasoning,
- experimental design,
- architectural decisions,
- observed behaviors,
- lessons learned,
- future directions.

3. Failure Pattern Database
File:
logs/failure_patterns.md

Document recurring failure modes and mitigation ideas.

4. Hypothesis Memory

File:
memory/hypotheses.md

Track active hypotheses.

5. Successful Patterns

File:
memory/successful_patterns.md

Document strategies that consistently work.

6. Rejected Ideas

File:
memory/rejected_ideas.md

Document abandoned or failed approaches.

Meta-Reasoning Requirements

The AI system must periodically perform self-analysis.

File:
logs/meta_reviews.md

These reviews should summarize:

- long-term trends,
- architectural evolution,
- recurring bottlenecks,
- strategic shifts,
- insights about ARC reasoning.

Git and Version Control

The AI system is expected to autonomously manage version control.

The system should create commits when:

- performance improves,
- architecture families change,
- parameter count decreases significantly,
- major hypotheses emerge,
- important infrastructure changes occur.

Commit messages should be descriptive and research-oriented.

Experimental Philosophy

The project should not optimize only for leaderboard score.

The research studies:

- autonomous reasoning,
- architecture discovery,
- iterative research behavior,
- compact neural program synthesis.

Failure analysis is as important as success.

The AI system should document:

- dead ends,
- repeated mistakes,
- unstable reasoning,
- overengineering tendencies,
- architecture collapse,
- ineffective search strategies.

Constraints

The AI system must obey all NeuroGolf competition constraints:

- valid ONNX outputs,

- static tensor shapes,
- operator restrictions,
- file size limits,
- competition rules.

The system should avoid disallowed operators and prioritize correctness and compactness simultaneously.

Research Priorities

Priority order:

1. Functional correctness
2. Generalization
3. Compression
4. Simplicity
5. Architectural elegance

Smaller and more interpretable solutions are preferred over large brute-force networks.

Long-Term Goal

The long-term goal is to investigate whether autonomous AI systems can:

- independently conduct ML research,
- iteratively improve neural architectures,
- discover compact reasoning structures,
- exhibit emergent research behavior under competitive constraints.

The AI system is not merely generating code.

It is participating in an autonomous AI research experiment.